Color Factor Evaluation: Every Step

From $\frac{1}{24}\sum_{a,b}\mathrm{Tr}[\lambda^a\lambda^b\lambda^b\lambda^a]$ to $\frac{32}{9}$

All Identities Sourced

PHYS 7502 — Parker Lewis

1 Starting Point

After squaring, using Hermiticity (Thomson Eq. 10.36), summing over quark colors, and averaging, we showed:

$$\mathcal{C} = \frac{1}{24}\sum_{a=1}^{8}\sum_{b=1}^{8}\mathrm{Tr}\Big[\lambda^a\lambda^b\lambda^b\lambda^a\Big] \tag{1}$$

We now evaluate this using three identities from the sources:

The Three Identities We Will Use

1. **Griffiths Problem 9.9** (the Fierz identity):

$$\sum_{a=1}^{8}(\lambda^a)_{ij}(\lambda^a)_{kl} = 2\delta_{il}\delta_{jk} - \frac{2}{3}\delta_{ij}\delta_{kl} \tag{2}$$

2. **Griffiths Eq. 9.26** (trace orthogonality, also Griffiths Problem 9.3):

$$\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab} \tag{3}$$

3. **General trace properties** (Griffiths Sec. 7.7, items 1–3):

$$\mathrm{Tr}(A+B) = \mathrm{Tr}(A) + \mathrm{Tr}(B) \tag{4}$$
$$\mathrm{Tr}(aA) = a\,\mathrm{Tr}(A) \quad (a = \text{scalar}) \tag{5}$$
$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA) \quad \text{(cyclic property)} \tag{6}$$

2 Step 1: Pull the Sum Over b Inside the Trace

The trace is a sum of diagonal elements. The sum $\sum_b$ only acts on the $\lambda^b\lambda^b$ part, while the λ^a matrices don't depend on b. Using linearity of the trace (4):

$$\sum_{b=1}^{8}\mathrm{Tr}\Big[\lambda^a\lambda^b\lambda^b\lambda^a\Big] = \mathrm{Tr}\left[\lambda^a\left(\sum_{b=1}^{8}\lambda^b\lambda^b\right)\lambda^a\right] \tag{7}$$

This is valid because $\mathrm{Tr}[A_1 + A_2 + \cdots] = \mathrm{Tr}[A_1] + \mathrm{Tr}[A_2] + \cdots$, and the λ^a matrices sandwiching the sum are the same in every term.

3 Step 2: Evaluate $\sum_b \lambda^b \lambda^b$ Using the Fierz Identity

This is the key step. We need the (m, n) element of the 3×3 matrix $\sum_b \lambda^b \lambda^b$:

$$\left(\sum_{b=1}^{8} \lambda^b \lambda^b \right)_{mn} = \sum_{b=1}^{8} \sum_{p=1}^{3} (\lambda^b)_{mp} \, (\lambda^b)_{pn} \tag{8}$$

The inner product $\sum_b (\lambda^b)_{mp} (\lambda^b)_{pn}$ is exactly the Fierz identity (2) with the index identifications $i = m$, $j = p$, $k = p$, $l = n$:

$$\sum_{b=1}^{8} (\lambda^b)_{mp} \, (\lambda^b)_{pn} = 2\delta_{mn}\delta_{pp} - \frac{2}{3}\delta_{mp}\delta_{pn} \tag{9}$$

Now sum over $p = 1, 2, 3$ (the repeated index from the matrix product):

$$\sum_{p=1}^{3} \left[2\delta_{mn}\delta_{pp} - \frac{2}{3}\delta_{mp}\delta_{pn} \right] \tag{10}$$

Evaluate each piece:

- **First term:** $\displaystyle\sum_{p=1}^{3} 2\delta_{mn}\delta_{pp} = 2\delta_{mn} \sum_{p=1}^{3} \delta_{pp} = 2\delta_{mn} \times 3 = 6\delta_{mn}$

  (Here $\delta_{pp} = 1$ for all p, and there are 3 values of p.)

- **Second term:** $\displaystyle\sum_{p=1}^{3} \frac{2}{3}\delta_{mp}\delta_{pn} = \frac{2}{3}\delta_{mn}$

  (The sum $\sum_p \delta_{mp}\delta_{pn} = \delta_{mn}$ — this is just matrix multiplication of two identity matrices.)

Combining:

$$\left(\sum_{b} \lambda^b \lambda^b \right)_{mn} = 6\delta_{mn} - \frac{2}{3}\delta_{mn} = \frac{18 - 2}{3}\delta_{mn} = \frac{16}{3}\delta_{mn} \tag{11}$$

Since δ_{mn} is the identity matrix:

Result (derived from Griffiths Problem 9.9)

$$\boxed{\sum_{b=1}^{8} \lambda^b \lambda^b = \frac{16}{3}\,\mathbb{I}_3} \tag{12}$$

This is $\frac{16}{3}$ times the 3×3 identity matrix. It is a direct consequence of the Fierz identity.

Explicit Verification Against Griffiths Examples

We can cross-check (12) against results Griffiths already computed:
Check 1: Griffiths Problem 9.9(a) asks to verify the Fierz identity for $i = j = k = l = 1$. That case gives $\sum_a (\lambda^a)_{11}(\lambda^a)_{11} = 2 \cdot 1 \cdot 1 - \frac{2}{3} \cdot 1 \cdot 1 = \frac{4}{3}$, and refers to Eq. 9.36 (the rr sextet color factor calculation), where Griffiths computes $(\lambda^3_{11})^2 + (\lambda^8_{11})^2 = 1 + 1/3 = 4/3$. ✓
Check 2: The diagonal element $(\sum_b \lambda^b \lambda^b)_{11} = \sum_b \sum_k |(\lambda^b)_{1k}|^2$ should be 16/3. Going through each λ^b:

λ^b	nonzero $(\lambda^b)_{1k}$	$\lvert(\lambda^b)_{1k}\rvert^2$
λ^1	$(\lambda^1)_{12} = 1$	1
λ^2	$(\lambda^2)_{12} = -i$	1
λ^3	$(\lambda^3)_{11} = 1$	1
λ^4	$(\lambda^4)_{13} = 1$	1
λ^5	$(\lambda^5)_{13} = -i$	1
λ^6	—	0
λ^7	—	0
λ^8	$(\lambda^8)_{11} = 1/\sqrt{3}$	1/3
	Total:	$5 + 1/3 = 16/3$ ✓

(The λ matrix entries are from Griffiths Eq. 9.9.)

4 Step 3: Substitute Back Into the Trace

Substituting (12) into (7):

$$\sum_b \mathrm{Tr}\Big[\lambda^a \lambda^b \lambda^b \lambda^a\Big] = \mathrm{Tr}\Big[\lambda^a \cdot \frac{16}{3}\,\mathbb{I}_3 \cdot \lambda^a\Big] \tag{13}$$

Now $\frac{16}{3}\,\mathbb{I}_3$ is a scalar (16/3) times the identity matrix. For any matrix A:

$$A \cdot (c\,\mathbb{I}) \cdot B = c\,(A \cdot B) \tag{14}$$

because the identity matrix does nothing. So:

$$\lambda^a \cdot \frac{16}{3}\,\mathbb{I}_3 \cdot \lambda^a = \frac{16}{3}\,\lambda^a \lambda^a \tag{15}$$

Using the linearity of the trace (Griffiths Sec. 7.7, item 2: $\mathrm{Tr}(aA) = a\,\mathrm{Tr}(A)$):

$$\mathrm{Tr}\Big[\frac{16}{3}\,\lambda^a \lambda^a\Big] = \frac{16}{3}\,\mathrm{Tr}[\lambda^a \lambda^a] \tag{16}$$

Substituting into (1):

$$\mathcal{C} = \frac{1}{24} \sum_{a=1}^{8} \frac{16}{3}\,\mathrm{Tr}(\lambda^a \lambda^a) \tag{17}$$

The factor 16/3 doesn't depend on a, so pull it out of the sum:

$$\mathcal{C} = \frac{1}{24} \cdot \frac{16}{3} \sum_{a=1}^{8} \mathrm{Tr}(\lambda^a \lambda^a) = \frac{16}{72} \sum_{a=1}^{8} \mathrm{Tr}(\lambda^a \lambda^a) \tag{18}$$

5 Step 4: Evaluate $\mathrm{Tr}(\lambda^a\lambda^a)$ and Sum Over a

From the trace orthogonality identity (3), setting $b = a$:

$$\mathrm{Tr}(\lambda^a\lambda^a) = 2\delta^{aa} = 2 \times 1 = 2 \qquad \text{(no sum on } a\text{)} \tag{19}$$

This holds for *every* a. Griffiths confirms this in Eq. 9.27: $\sum_a \mathrm{Tr}(\lambda^a\lambda^a) = 16$, which is just 2×8.

Explicit Verification of Griffiths Eq. 9.26

Let's check a few cases by direct matrix multiplication.

$a = 1$**:**

$$\lambda^1\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{20}$$

$\mathrm{Tr}(\lambda^1\lambda^1) = 1 + 1 + 0 = 2$ ✓

$a = 2$**:**

$$\lambda^2\lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} (-i)(i) & 0 & 0 \\ 0 & (i)(-i) & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{21}$$

$\mathrm{Tr}(\lambda^2\lambda^2) = 1 + 1 + 0 = 2$ ✓

$a = 3$**:**

$$\lambda^3\lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{22}$$

$\mathrm{Tr}(\lambda^3\lambda^3) = 1 + 1 + 0 = 2$ ✓

$a = 8$**:**

$$\lambda^8\lambda^8 = \frac{1}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \frac{1}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \tag{23}$$

$\mathrm{Tr}(\lambda^8\lambda^8) = \frac{1}{3}(1 + 1 + 4) = \frac{6}{3} = 2$ ✓

The pattern is the same for all 8 matrices (the normalization $\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$ was chosen precisely to give this).

Now sum over all 8 values of a:

$$\sum_{a=1}^{8} \mathrm{Tr}(\lambda^a\lambda^a) = \underbrace{2 + 2 + 2 + 2 + 2 + 2 + 2 + 2}_{8 \text{ terms}} = 16 \tag{24}$$

This is Griffiths Eq. 9.27.

6 Step 5: Final Arithmetic

Substituting (24) into (18):

$$\mathcal{C} = \frac{16}{72} \times 16 \tag{25}$$

$$= \frac{16 \times 16}{72} \tag{26}$$

$$= \frac{256}{72} \tag{27}$$

Simplify by dividing numerator and denominator by $\gcd(256, 72) = 8$:

$$\frac{256}{72} = \frac{256 \boldsymbol{\nabla} \cdot 8}{72 \boldsymbol{\nabla} \cdot 8} = \frac{32}{9} \tag{28}$$

Color Factor

$$\boxed{\mathcal{C} = \frac{32}{9}} \tag{29}$$

Where each number came from:

$$\mathcal{C} = \underbrace{\frac{1}{24}}_{\substack{\text{avg: } \frac{1}{3}\times\frac{1}{8} \\ \text{(Thomson 10.28)}}} \times \underbrace{\frac{16}{3}}_{\substack{\sum_b \lambda^b\lambda^b = \frac{16}{3}\mathbb{I} \\ \text{(from Griffiths Prob. 9.9)}}} \times \underbrace{16}_{\substack{\sum_a \mathrm{Tr}(\lambda^a\lambda^a) \\ \text{(Griffiths Eq. 9.27)}}}$$

Cross-check against Thomson Sec. 10.7.2: Thomson computes the $qq \to qq$ color factor using $\langle|C|^2\rangle = \frac{1}{144}\sum_{a,b}[\mathrm{Tr}(\lambda^a\lambda^b)]^2 = \frac{1}{144}\sum_a(2)^2 = \frac{4\times8}{144} = \frac{2}{9}$. Our problem is different ($qg \to qg$ has *two* λ matrices per vertex instead of one), but the method — reduce to traces, apply $\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$ — is the same.

7 Source Index

Identity used	Source	Where applied
$\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$	Griffiths Eq. 9.26 / Prob. 9.3	Step 4
$\sum_a \mathrm{Tr}(\lambda^a\lambda^a) = 16$	Griffiths Eq. 9.27	Step 4
Fierz: $\sum_a(\lambda^a)_{ij}(\lambda^a)_{kl} = 2\delta_{il}\delta_{jk} - \frac{2}{3}\delta_{ij}\delta_{kl}$	Griffiths Prob. 9.9	Step 2
$\lambda^{a\dagger} = \lambda^a$ (Hermiticity)	Griffiths Eq. 9.9	Squaring step
$CC^* = \lambda^a_{ji}(\lambda^b_{ji})^* = \lambda^a_{ji}\lambda^b_{ij}$	Thomson Eq. 10.36	Squaring step
$\mathrm{Tr}(aA) = a\,\mathrm{Tr}(A)$	Griffiths Sec. 7.7, item 2	Step 3
$\mathrm{Tr}(A+B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$	Griffiths Sec. 7.7, item 1	Step 1
$\langle\|C\|^2\rangle = \frac{1}{N_{\text{init}}}\sum\|C\|^2$	Thomson Eq. 10.28–10.29	Averaging
Gell-Mann matrices (explicit form)	Griffiths Eq. 9.9	Verification